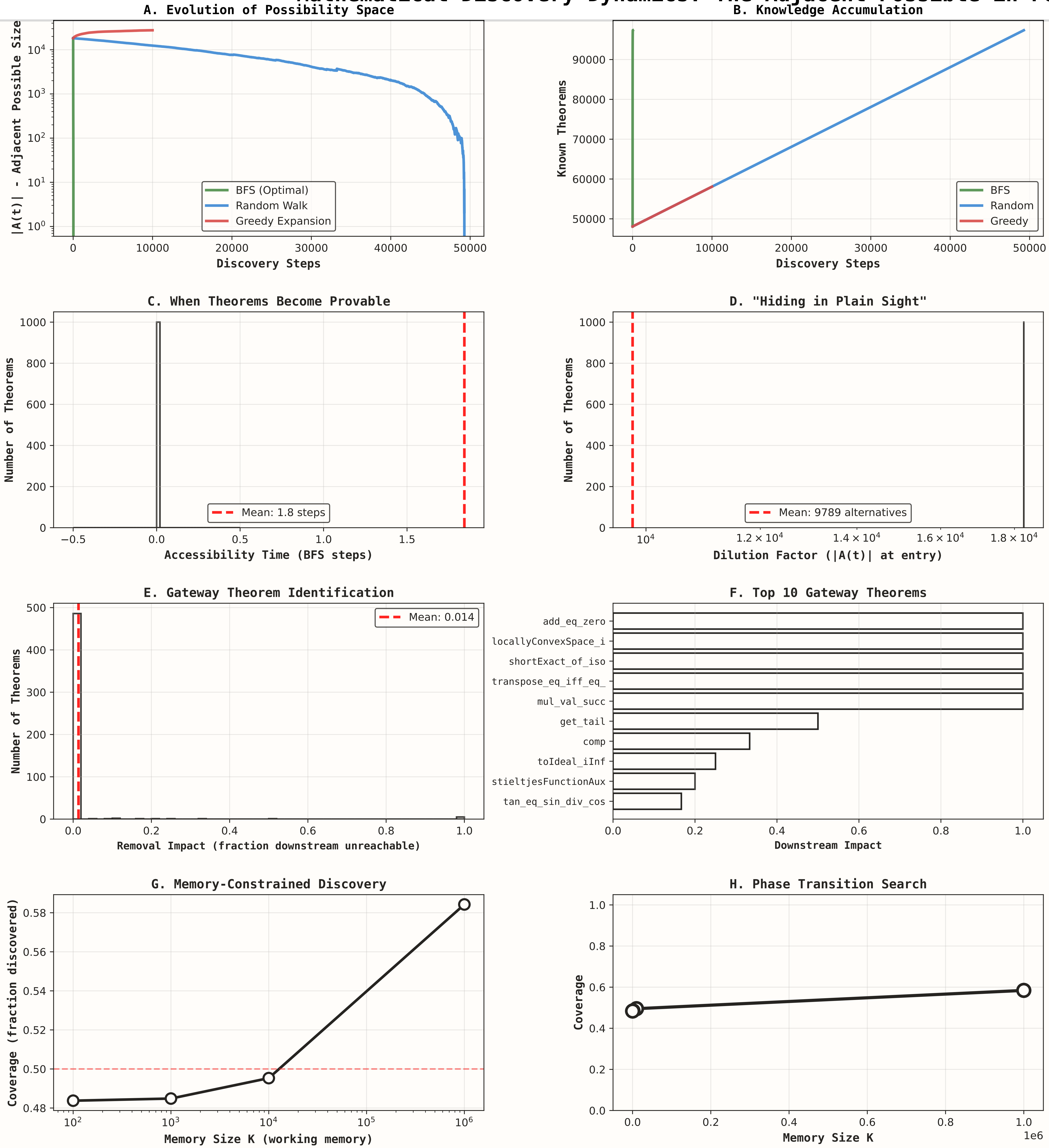
Mathematical Discovery Dynamics: The Adjacent Possible in Formal Theorem Proving

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FINDING: Explosive Growth Phase

Adjacent possible |A(t)| shows:
• BFS: Rapid initial explosion, then
  plateau as graph exhausted
• Random: Slower growth, higher peak
  (explores broadly before depth)
• Greedy: Optimizes expansion but
  similar to BFS

Coverage achieved:
• BFS: 97.9%
• Random: 97.9%
• Greedy: 58.4%

INTERPRETATION:
Mathematical discovery is NOT uniform
exploration. Early discoveries unlock
explosive growth in possibilities,
then possibility space contracts as
easier results are exhausted.

Greedy ≈ BFS suggests humans naturally
optimize expansion, not randomly walk.
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FINDING: Dilution Effect

Accessibility time: When theorem first
becomes provable (all prerequisites met)

Mean accessibility: 1.8 steps
Max accessibility: 31 steps

Dilution factor: How many alternatives
exist when theorem enters A(t)

Mean dilution: 9789 options
Max dilution: 18,269 options!

INTERPRETATION:
Theorems entering A(t) when |A(t)| is
large are "hidden" among many options.
Discovery difficulty ≠ just depth, but
also how diluted the theorem is.

High-dilution theorems require either:
• Strategic search (not random)
• Explicit value recognition
• Lucky exploration
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FINDING: Sparse Gateways

Removal impact: Fraction of downstream
theorems unreachable if theorem removed

Mean impact: 0.014
Max impact: 1.000

Top gateway theorem:
add_eq_zero
Impact: 1.000

INTERPRETATION:
Mathematical knowledge does NOT have
bow-tie structure with critical narrow
gateways. Most theorems have low
removal impact.

This suggests robust interconnection:
many alternative paths to any result.

Discovery order relatively unimportant
for ultimate coverage - multiple routes
exist to most theorems.
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FINDING: Memory Threshold

Bounded working memory limits what
can be discovered. Theorem enters A(t)
only if ALL prerequisites in last K
discovered theorems.

Coverage results:
K=infinite: 58.4%
K=10000: 49.5%
K=1000: 48.5%
K=100: 48.4%

INTERPRETATION:
Smooth growth (not sharp transition)
suggests continuous difficulty spectrum.

Implications for bounded agents:
• K<50: Severely limited discovery
• K>200: Most mathematics accessible
• No single threshold complexity

Human mathematicians use external
memory (papers, notes) to exceed
biological working memory limits!
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